

Data Management Technologies

Unit 9 Consolidation Task

Activity 1. Concept Review Questions

1). Define the **key components of transaction management and explain their importance.**

(A transaction is a logical collection of operations that are undertaken on a database in order to perform a practical task, such as adding new users, or updating account balances).

Components of transaction management include:

- **ACID (Atomicity, Consistency, Isolation, and Durability)**
 - **Atomicity:** Each transaction is taken as an undivided whole and is either fully successful or fully unsuccessful.
 - **Consistency:** The database should remain in a valid state before and after the transactions. Parameters that are not intended to be changed should remain consistent
 - **Isolation:** Transactions should be able to run independently
 - **Durability:** Data stored in the system should be persistent, even when the system fails. This can be ensured through regular backups and the use of non-volatile storage

2). Describe the **challenges associated with concurrency control in database systems.**

Concurrency control is a mechanism for DBMS to simultaneously perform transactions, while maintaining the ACID properties.

- Transaction Neglect: Some transactions might be continuously postponed and never completed.
- Deadlocks: This occurs when two transactions require a resource held by the other. In this scenario, the transactions cannot complete concurrently until the deadlock is resolved.
- Lost updates: This occurs when multiple transactions update the same data concurrently, leading to versions of the update being overwritten and lost.

3). Explain how **database security mechanisms protect sensitive data.**

- **Authentication:** Verifying the identity of users seeking to access the data
- **Authorisation:** ensuring that users have the authority to perform the actions they are seeking to
- **Backups** – Mitigates ransomware attacks by providing additional copies of data.
- **Logging and Review of data access:** This helps to detect and penalised unauthorized access and access attempts.

References

IBM. (2002). *Concurrency control*. IBM Documentation.

<https://www.ibm.com/docs/en/was/8.5.5?topic=beans-concurrency-control>